

Track 17: Prompt Engineering

June 2026

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MR ROKESH TECHNOLOGY — Training & Internship Programs

Skill path: LLM Fundamentals → Prompt Patterns → Chain-of-Thought → RAG → Evaluation → Production APIs

Related: [Track 18 — AI Agent Development](#) | [Track 34 — Generative AI for Professionals](#) | [Track 33 — Prompt Engineering for Business](#)

Track Overview

Who this is for: Developers, AI engineers, and technical professionals building LLM-powered applications.

Prerequisites: Basic Python; familiarity with APIs helpful. For non-technical audience see [Track 33](#).

Tools: OpenAI API, Anthropic API, LangChain, LlamaIndex, Python, Jupyter, vector DBs (Chroma/Pinecone), prompt evaluation frameworks.

Certification targets: Portfolio of production prompt systems (9-month)

Outcomes by Duration

Duration	Hours	Job Readiness	Key Deliverables
2 Weeks	40–50	Prompt literacy	Prompt library + API demo app
1 Month	80–100	Junior prompt engineer	RAG mini-app
3 Months	250–300	Internship-ready	Production prompt pipeline capstone
6 Months	500–600	AI app developer	Multi-model prompt system
9 Months	750–900	AI engineer track	Agent-ready prompt framework + 2 capstones

2-Week Crash Course

Mini Project: Prompt library + API chatbot | **Level:** L1

Days	Topics	Deliverable
D1–2	LLM fundamentals, tokens, temperature, system prompts	Prompt cheat sheet
D3–4	Zero-shot, few-shot, role prompting	20 tested prompt templates
D5–6	Chain-of-thought, structured output (JSON)	Structured extraction demo
D7–8	OpenAI/Anthropic API integration	Python chatbot script
D9–10	Evaluation basics, iteration	Documented prompt library + demo

1-Month Foundation

Week	Topics	Project
W1	Prompt patterns, persona, constraints	30-pattern prompt pack
W2	Embeddings, vector search intro	Semantic search demo
W3	RAG architecture, chunking strategies	Document Q&A app
W4	Evaluation metrics, A/B testing prompts	RAG mini-app with eval report

3-Month Job Ready

Month	Topics	Projects
M1	Advanced prompting, function calling, tools	Tool-use chatbot
M2	LangChain/LlamaIndex, RAG pipelines	Multi-doc knowledge base
M3	Guardrails, cost optimization, capstone	Production prompt + RAG system

Capstone: Ingest PDF corpus; accurate Q&A; latency/cost report; failure case documentation.

6-Month Professional

Month	Topics	Projects
M1–M3	As 3-month track	—
M4	Multi-model routing, fallback strategies	Model router service
M5	Fine-tuning intro, prompt versioning	Versioned prompt registry
M6	Production deployment, monitoring	Deployed LLM application

9-Month Advanced

Capstones: Enterprise RAG platform + eval-driven prompt optimization framework | **Level:** L5

Additional: Adversarial prompt testing, red teaming, multi-agent prompt orchestration (bridge to [Track 18](#)).

Assessment Rubric

Component	Weight	Criteria
Weekly Quiz	20%	LLM concepts, prompt patterns, RAG
Lab Completion	40%	Prompts tested; APIs functional
Project	30%	Accuracy, cost, documentation
Portfolio	10%	GitHub repos, prompt libraries

Appendix: Mentor Notes

Distinction: Track 17 is **developer/technical**. Track 33 is **business/non-code**. Track 34 is **professional productivity** across roles.